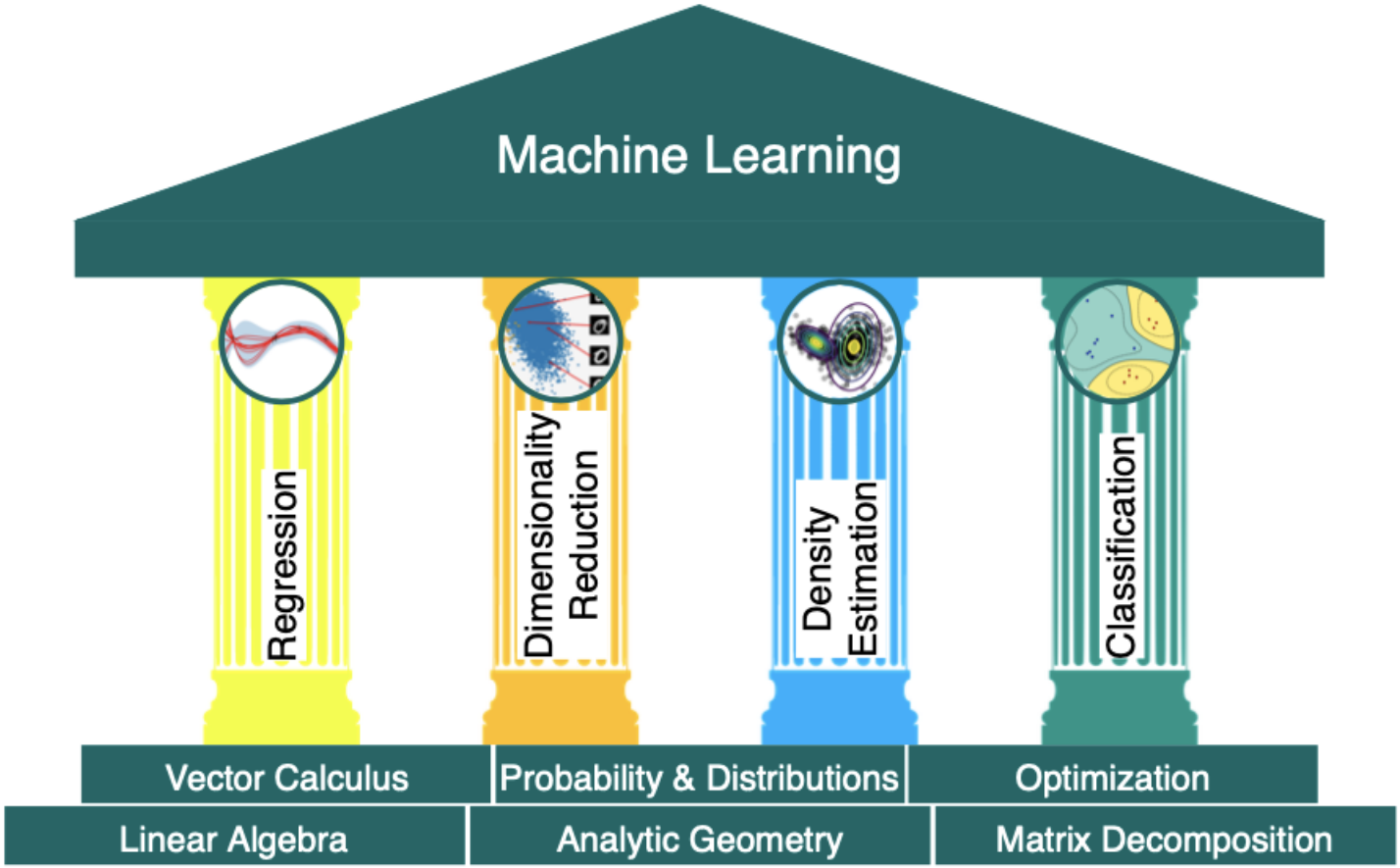


Figure: The foundations and four pillars of machine learning.

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What is Linear Algebra?

Multidimensional mathematical objects-like vectors and matrices, and operations on them.

Scalars

A scalar is just a number (8,10.2, π, etc.). We usually use the notation x to refer to this variable.

Vectors

Un vecteur est un tableau 1D de nombres ayant une orientation horizontale ou verticale.

Row vector

Horizontal $\Longrightarrow \mathbf{x} = [x_1 \ x_2 \ x_3]$, this is known as *row vector* composed of 3 elements.

Column vector

Vertical $\Longrightarrow$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

, this is known as *column vector* composed of 4 elements.

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System of linear equations

Many problems can be formulated as systems of linear equations, and linear algebra gives us the tools for solving them.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

We aim at solving the system $\Longrightarrow$ find the values of the *unknowns* $x_1, x_2, \ldots, x_n$.

Example

Let x_1, x_2 and x_3 be 3 unknowns such that:

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 9 && (1)\\ -x_1 + 3x_2 - x_3 &= -6 && (2)\\ 2x_1 - 5x_2 + 5x_3 &= 17 && (3) \end{aligned}$$

$(1) + (2) \implies x_2 + 2x_3 = 3 \qquad (4)$

Multiply (1) by -2 and add the resulting equation to (3)

$\implies -x_2 - x_3 = -1 \qquad (5)$

$(4) + (5) \implies \boxed{x_3 = 2}$

Replace the value of x_3 in (4) $\implies \boxed{x_2 = -1}$

Substitute x_2 and x_3 into (1) $\implies \boxed{x_1 = 1}$

From system to matrices

The above example can be simplifies by collecting the *coefficients* into vectors :

$$\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} -2 \\ 3 \\ -5 \end{bmatrix} x_2 + \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix} x_3 = \begin{bmatrix} 9 \\ -6 \\ 17 \end{bmatrix}$$

Those vectors can be also collected into matrices:

$$\begin{bmatrix} 1 & -2 & 3 \\ -1 & 3 & -1 \\ 2 & -5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 17 \end{bmatrix}$$

The matricial form of any linear system of n unknowns and m equations can be expressed as:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \Leftrightarrow \mathbf{Ax} = \mathbf{b}$$

Matrices

A matrix is a rectangular array of numbers arranged in rows and columns such that its general term a_{ij} identifies that the element belongs to the $i-th$ row and $j-th$ column.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Matrices are used in/for:

- Linear Algebra: represent and solve systems of linear equations, eigenvalue and eigenvector computations, etc.
- Computer Science: image processing, data manipulation
- Statistics and Data Analysis: Regression,

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Addition and Subtraction

The matrices should have the **same dimensions (n×n)**.

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

$$A + B = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}$$

$$A - B = \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} \\ a_{21} - b_{21} & a_{22} - b_{22} \end{pmatrix}$$

Example

$$A = \begin{pmatrix} 2 & 5 & -3 \\ 4 & -1 & 7 \\ 6 & 0 & 8 \end{pmatrix} \quad B = \begin{pmatrix} -1 & 3 & 4 \\ 5 & -2 & 0 \\ 7 & 2 & -6 \end{pmatrix}$$

$$A+B = \begin{pmatrix} 2+(-1) & 5+3 & -3+4 \\ 4+5 & -1+(-2) & 7+0 \\ 6+7 & 0+2 & 8+(-6) \end{pmatrix} = \begin{pmatrix} 1 & 8 & 1 \\ 9 & -3 & 7 \\ 13 & 2 & 2 \end{pmatrix}$$

$$A-B = \begin{pmatrix} 2-(-1) & 5-3 & -3-4 \\ 4-5 & -1-(-2) & 7-0 \\ 6-7 & 0-2 & 8-(-6) \end{pmatrix} = \begin{pmatrix} 3 & 2 & -7 \\ -1 & 1 & 7 \\ -1 & -2 & 14 \end{pmatrix}$$

Multiplication

Verify that the matrices are compatible

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

Multiply each row by each column:

$$A \times B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

Remark

The second dimension of the first matrix should be equal to the first dimension of the second matrix. $A(m \times n)$ and $B(n \times d)$

Example

$$A = \begin{pmatrix} 2 & 15 & 8 \\ 5 & 3 & 12 \\ 11 & 7 & 9 \\ 14 & 1 & 6 \end{pmatrix} \quad B = \begin{pmatrix} 10 & 4 \\ 9 & 16 \\ 5 & 13 \end{pmatrix}$$

$$C = A \times B = \begin{pmatrix} 2\cdot 10+15\cdot 9+8\cdot 5 & 2\cdot 4+15\cdot 16+8\cdot 13 \\ 5\cdot 10+3\cdot 9+12\cdot 5 & 5\cdot 4+3\cdot 16+12\cdot 13 \\ 11\cdot 10+7\cdot 9+9\cdot 5 & 11\cdot 4+7\cdot 16+9\cdot 13 \\ 14\cdot 10+1\cdot 9+6\cdot 5 & 14\cdot 4+1\cdot 16+6\cdot 13 \end{pmatrix}$$

$$= \begin{pmatrix} 195 & 352 \\ 137 & 224 \\ 218 & 273 \\ 179 & 150 \end{pmatrix}$$

Scalar multiplication

let λ be a scalar and $\mathbf{A}$ be a matrix (can be of any dimension).

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{pmatrix}$$

The scalar multiplication of λ and $\mathbf{A}$ is given by:

$$\lambda\mathbf{A} = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \lambda a_{13} \\ \lambda a_{21} & \lambda a_{22} & \lambda a_{23} \\ \lambda a_{31} & \lambda a_{32} & \lambda a_{33} \\ \lambda a_{41} & \lambda a_{42} & \lambda a_{43} \end{pmatrix}$$

Example

$$\lambda\mathbf{A} = 3\begin{pmatrix} 2 & 5 & -3 \\ 4 & -1 & 7 \end{pmatrix} = \begin{pmatrix} 3\cdot 2 & 3\cdot 5 & 3\cdot -3 \\ 3\cdot 4 & 3\cdot -1 & 3\cdot 7 \end{pmatrix} = \begin{pmatrix} 6 & 15 & -9 \\ 12 & -3 & 21 \end{pmatrix}$$

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Determinants

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The determinant of A is given by:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

The determinant of B is:

$$\det(A) = (1 \cdot 4) - (2 \cdot 3) = 4 - 6 = -2$$

For a 3×3 matrix:

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

The determinant of matrix A is given by:

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{1j})$$

$$\det(A) = \begin{matrix} + & - & + \\ \end{matrix} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Which simplifies to:

$$\begin{aligned} \det(A) = a_{11}(a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) - a_{12}(a_{21} \cdot a_{33} - a_{23} \cdot a_{31}) \\ + a_{13}(a_{21} \cdot a_{32} - a_{22} \cdot a_{31}) \end{aligned}$$

Consider the matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

The determinant of A is:

$$\det(A) = 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

Calculating the 2×2 determinants:

$$\det(A) = 1(5 \cdot 9 - 6 \cdot 8) - 2(4 \cdot 9 - 6 \cdot 7) + 3(4 \cdot 8 - 5 \cdot 7)$$

$$\det(A) = 1(45 - 48) - 2(36 - 42) + 3(32 - 35)$$

$$\det(A) = 1(-3) - 2(-6) + 3(-3)$$

$$\det(A) = -3 + 12 - 9 = 0$$

Inverses

For any scalar $x \neq 0$, $\exists$ another scalar y such that $xy = 1$. In this case, $y = \frac{1}{x} = x^{-1}$

Consider a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$. By analogy, we wonder that since we have an identity matrix $\mathbf{I_n}$ that acts like the number 1 (as above), then $\exists$ another square matrix $\mathbf{B}$, such that:

$$\mathbf{AB} = \mathbf{I_n} = \mathbf{BA}$$

, where

$$\mathbf{I_n} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$\mathbf{B}$ is called the *inverse matrix* of $\mathbf{A}$ and it is denoted by $\mathbf{A}^{-1}$. In this case, we say that $\mathbf{A}$ is *invetible*, *regular*, or *nonsingular*.

However, it is not guaranteed that every matrix $\mathbf{A}$ is invertible. To check if A has an inverse, we calculate its determinant.
If $\det(\mathbf{A}) = 0$, then $\mathbf{A}$ has no inverse. In this case, we say that $\mathbf{A}$ is *noninvertible* or *singular*.

How to find the Inverse

- Gaussian elimination (Row echelon form)
- Adjoint (Transpose of Co-factor) matrix

Consider the matrix A:

$$A = \begin{bmatrix} 4 & 3 & 2 \\ 3 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

Adjoint (Adjugate) matrix

To find the matrix of minors, we calculate the determinant of the 2×2 matrix that remains after removing the row and column of each element. The matrix of minors for A is:

$$\text{Minors} = \begin{bmatrix} \det\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix} & \det\begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix} & \det\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \\ \det\begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix} & \det\begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} & \det\begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix} \\ \det\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} & \det\begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} & \det\begin{pmatrix} 4 & 3 \\ 3 & 1 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 2 & 7 & 1 \\ 7 & 8 & -2 \\ 1 & -2 & -5 \end{bmatrix}$$

The matrix of cofactors is:

$$\text{Cofactors} = \begin{bmatrix} +2 & -7 & +1 \\ -7 & +8 & -(-2) \\ +1 & -(-2) & -5 \end{bmatrix} = \begin{bmatrix} 2 & -7 & 1 \\ -7 & 8 & +2 \\ 1 & 2 & +(-5) \end{bmatrix}$$

The adjugate of A is the transpose of the matrix of cofactors:

$$\text{adj}(A) = \begin{bmatrix} 2 & -7 & 1 \\ -7 & 8 & 2 \\ 1 & 2 & -5 \end{bmatrix}$$

The determinant of A is calculated as follows:

$$\det(A) = 4(1 \cdot 3 - 1 \cdot 1) - 3(3 \cdot 3 - 1 \cdot 2) + 2(3 \cdot 1 - 1 \cdot 2)$$

$$\det(A) = 4(2) - 3(7) + 2(1) = 8 - 21 + 2 = -11$$

Finally, the inverse of the matrix A is:

$$\mathbf{A}^{-1} = \frac{1}{\det(A)}\text{adj}(A) = \frac{1}{-11}\begin{bmatrix} 2 & -7 & 1 \\ -7 & 8 & 2 \\ 1 & 2 & -5 \end{bmatrix}$$

Transpose

Let the original matrix $\mathbf{A}$ be:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

The transpose of $\mathbf{A}$ is the matrix $\mathbf{A}^T$ obtained by flipping the rows and columns along the main diagonal.

$$\mathbf{A}^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{bmatrix}$$

Solving Linear Equations

As discussed earlier, when we solve a linear system, we seek the vector of unknowns $\mathbf{x}$ that it supposed to find one or more solutions (assume it exists) for a given system $\mathbf{Ax} = \mathbf{b}$.
To find $\mathbf{x}$, we should use find the inverse matrix $\mathbf{A}^{-1}$:

- If $\mathbf{A}$ is a square and invertible matrix, then

$$\mathbf{Ax} = \mathbf{b} \Leftrightarrow \mathbf{A}^{-1}\mathbf{Ax} = \mathbf{A}^{-1}\mathbf{b} \Leftrightarrow \boxed{\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}}$$

- Otherwise, under mild assumptions ($\mathbf{A}$ should have *linearly independent* columns),

$$\mathbf{Ax} = \mathbf{b} \Leftrightarrow \mathbf{A}^{\mathbf{T}}\mathbf{Ax} = \mathbf{A}^{\mathbf{T}}\mathbf{b} \Leftrightarrow \boxed{\mathbf{x} = (\mathbf{A}^{\mathbf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathbf{T}}\mathbf{b}}$$

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Groups

Consider a set $\mathscr{V}$ and an operation $\otimes$ defined on $\mathscr{V}$ such that:

$$\otimes : \mathscr{V} \times \mathscr{V} \mapsto \mathscr{V}.$$

$V = (\mathscr{V}, \otimes)$ is called a group if the following holds:

- **Closure**: For all $a, b \in \mathscr{V}$, the sum $a \otimes b \in \mathscr{V}$.
- **Associativity**: For all $a, b, c \in \mathscr{V}$, $(a \otimes b) \otimes c = a \otimes (b \otimes c)$.
- **Neutral element**: There exists an element $e \in \mathscr{V}$ such that for all $a \in \mathscr{V}$, $a \otimes e = e \otimes a = a$.
- **Inverse element**: For each $a \in \mathscr{V}$, there exists an element $b \in \mathscr{V}$ such that $a \otimes (b) = e$ and $b \otimes a = e$, where e the neutral element.

If for all $a, b \in \mathscr{V}$, $a \otimes b = b \otimes a$, then $V = (\mathscr{V}, \otimes)$ is an Abelian group (**commutative**).

Group: Set + inner operations on the set (mappings that only operate on elements in the set)

Vector Space

A real-valued *vector space* $V = (\mathscr{V}, +, .)$ is a set $\mathscr{V}$ with two operations

$$inner + \quad : \quad \mathscr{V} \times \mathscr{V} \mapsto \mathscr{V} \tag{4}$$

$$outer . \quad : \quad \mathbb{R} \times \mathscr{V} \mapsto \mathscr{V} \tag{5}$$

where

- $(\mathscr{V}, +)$ is an Abelian group ()
- Distributivity:
  - $\forall\ \alpha \in \mathbb{R},\ x,\ y \in \mathscr{V} : \alpha(x+y) = \alpha x + \alpha y$
  - $\forall\ \alpha, \beta \in \mathbb{R},\ x \in \mathscr{V} : (\alpha+\beta)x = \alpha x + \beta x$
- Associativity (outer operation): $\forall\ \alpha, \beta \in \mathbb{R},\ x \in \mathscr{V} : \alpha(\beta x) = (\alpha\beta)x$
- Neutral element with respect to the outer operation: $\forall\ x \in \mathscr{V} : 1.x = x$

Vector space: examples

- $\mathscr{V} = \mathbb{R}^n, n \in \mathbb{N}$ is a vector space with operations defined as follows:
  - Addition: $\mathbf{x} + \mathbf{y} = (x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n)$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
  - Multiplication by scalars: $\lambda \mathbf{x} = \lambda (x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n)$ for all $\lambda \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n$
- $\mathscr{V} = \mathbb{R}^{m \times n}, m, n \in \mathbb{N}$ is a vector space with
  - Addition: $\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{bmatrix}$ is defined elementwise for all $\mathbf{A}, \mathbf{B} \in \mathscr{V}$
  - Multiplication by scalars: $\lambda \mathbf{A} = \begin{bmatrix} \lambda a_{11} & \cdots & \lambda a_{1n} \\ \vdots & \ddots & \vdots \\ \lambda a_{m1} & \cdots & \lambda a_{mn} \end{bmatrix}$ as defined in $\mathbb{R}^{m \times n}$

Linear Combination

Consider a vector space V of finite number of vector $\mathbf{x_1}, \ldots, \mathbf{x_k} \in V$. Let $\lambda \in \mathbb{R}$. Then every $v \in V$ having the form

$$v = \sum_{i=1}^{k} \lambda_i \mathbf{x_i} = \lambda_1 \mathbf{x_1} + \cdots + \lambda_k \mathbf{x_k} \in V$$

is a *linear combination* of the vectors $\mathbf{x_1}, \ldots, \mathbf{x_k}$.

Remark

Linear combinations are usually introduced to find linear dependencies among vectors, calculate ranks and basis, etc (behind the scope of this course)

Linear Mappings

Let V and W be two vector spaces. A mapping $\Phi : V \mapsto W$ is called a *linear mapping* (linear transformation/ vector space homomorphism) if $\forall\, \mathbf{x}, \mathbf{y} \in V, \forall \boldsymbol{\lambda} \in \mathbb{R}$,

$$\begin{aligned} \boldsymbol{\Phi}(\mathbf{x}, \mathbf{y}) &= \boldsymbol{\Phi}(\mathbf{x}) + \boldsymbol{\Phi}(\mathbf{y}) \\ \boldsymbol{\Phi}(\boldsymbol{\lambda}\mathbf{x}) &= \boldsymbol{\lambda}(\boldsymbol{\Phi}(\mathbf{x})) \end{aligned}$$

Special Mappings

When a function is both injective and surjective, it is called bijective. This means that every element in the domain maps to a unique element in the codomain, and every element in the codomain is mapped by some element in the domain.

Injective (One-to-One)

A function $\Phi : A \to B$ is injective if different elements in the domain A map to different elements in the codomain B. Formally, Φ is injective if:

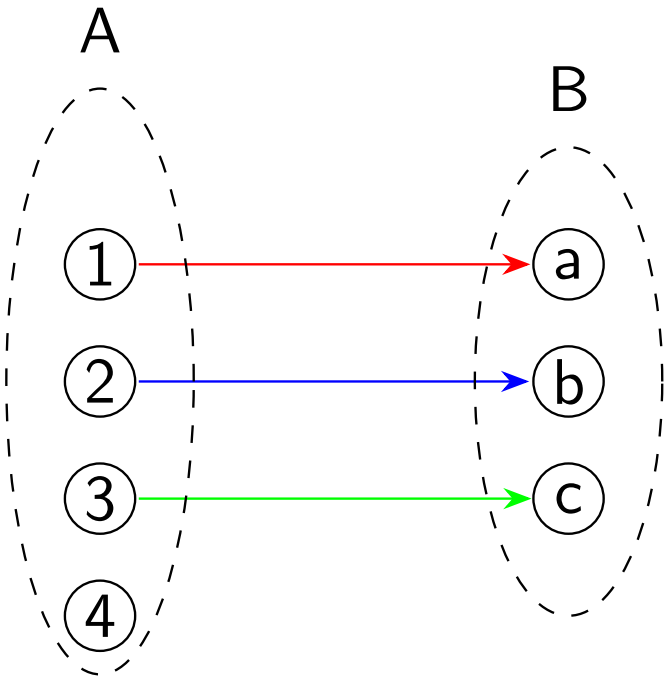
$$\Phi(a_1) = \Phi(a_2) \implies a_1 = a_2 \quad \text{for all } a_1, a_2 \in A$$

Surjective (One-to-One)

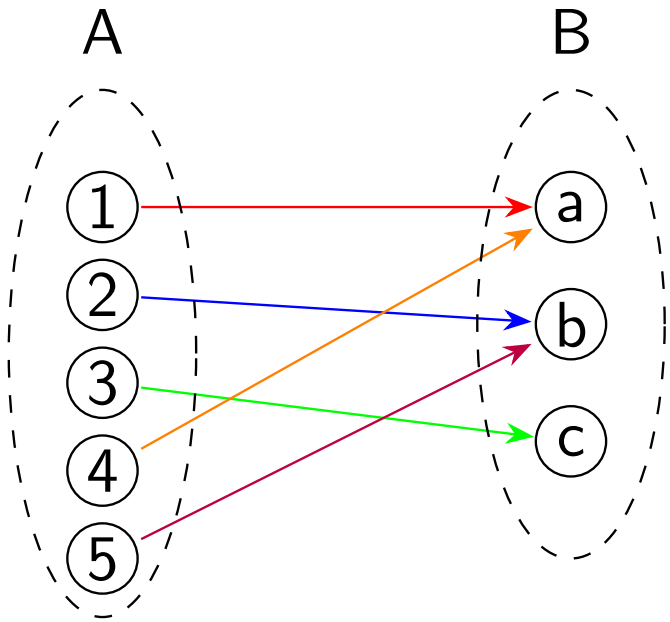
A function $\Phi : A \to B$ is surjective if every element in the codomain B is the image of at least one element in the domain A. Formally, Φ is surjective if:

$$\text{for every } b \in B, \text{ there exists an } a \in A \text{ such that } \Phi(a) = b \ (\Phi(A) = B)$$

Frame

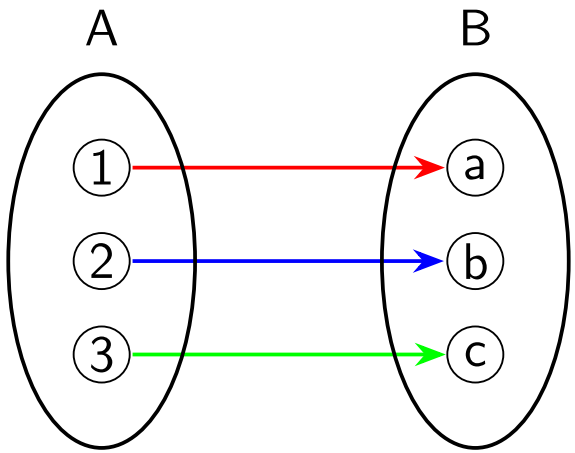


Injective



Surjective

Bijectivity



Bijection

Example

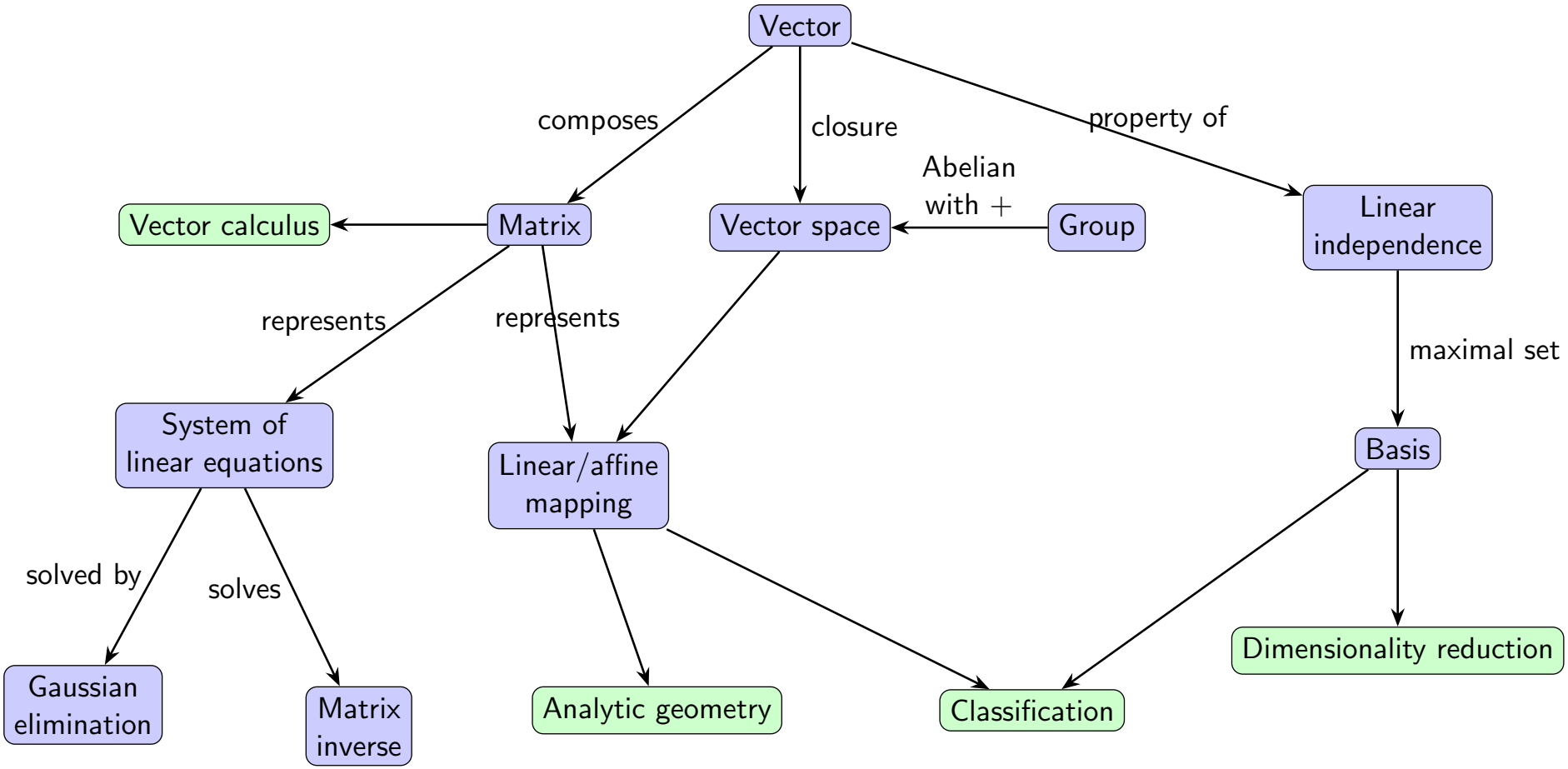
Consider the function $\Phi : A \to B$ defined by:

$$\Phi(1) = a, \quad \Phi(2) = b, \quad \Phi(3) = c$$

- **Injective**: Each element in A maps to a unique element in B.
- **Surjective**: Every element in B is mapped by some element in A.
- A mapping Φ has an inverse if and only if it is both surjective and injective.

Put it all together

relation between invertiblity and iterative methods



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Norms

Let V be a vector space. A **norm** on a vector space V is a function: $\|.\| : V \to \mathbb{R}$ satisfying the following conditions:

- positive definiteness: $\|\mathbf{v}\| \geq 0 \ \forall \ \mathbf{v} \in V$ and $\|\mathbf{v}\| = 0 \Leftrightarrow \mathbf{v} = 0$
- absolutely homogeneous: $\|\lambda \mathbf{v}\| = \|\lambda\| \|\mathbf{v}\|, \ \forall \ \mathbf{v} \in V$ and$\lambda \in \mathbb{R}$
- triangle inequality: $\|\mathbf{v} + \mathbf{w}\| = \|\mathbf{v}\| + \|\mathbf{w}\|, \ \forall \ \mathbf{v}, \mathbf{w} \in V$

If $\|.\|$ is a norm on V, then the pair $(V, \|.\|)$ is called a **normed vector space**.

Remark

A norm maps a vector $\mathbf{x}$ (real or complex) to a **non-negative** real number $x \in \mathbb{R}$, such that $x \geq 0$.

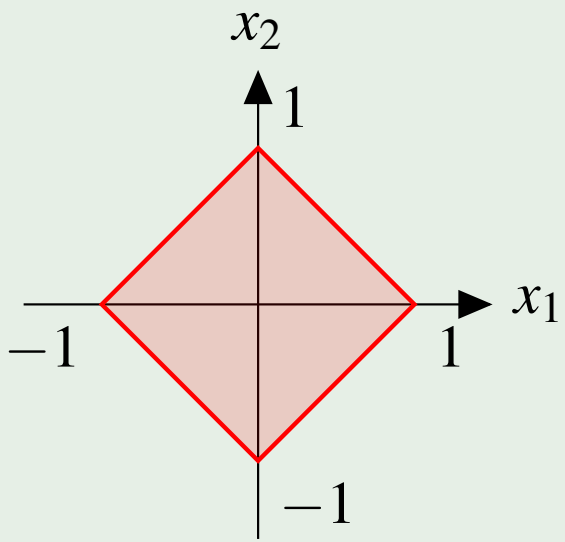
Some known norms

Manhattan Norm

Let $\mathbf{x} \in \mathbb{R}^n$. The Manhattan norm (or ℓ^1 norm) of a vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is the sum of the absolute values of its components:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

For example, if $\|\mathbf{x}\|_1 = 1$, the set of vectors with norm 1 are the red lines:

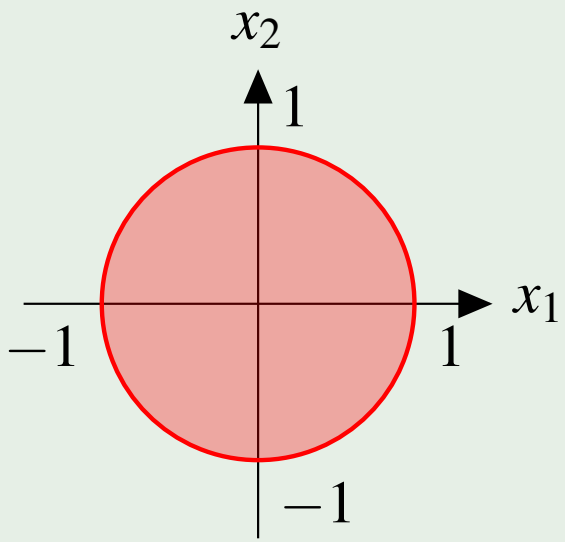


Euclidean Norm

The Euclidean norm (or ℓ^2 norm) of a vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is defined as:

$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} = \sqrt{\mathbf{x}^T \mathbf{x}} \tag{6}$$

If $\|\mathbf{x}\|_2 = 1$, then



Inner product

Let V be a vector space. An **inner product** on V is a rule that assigns to each pair $\mathbf{v}, \mathbf{w} \in V$ a real number $\langle \mathbf{v}, \mathbf{w} \rangle$ such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\alpha \in \mathbf{R}$,

- $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$,
- $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$,
- $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$,
- $\langle \alpha \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{v}, \mathbf{w} \rangle$.

If $\langle ., . \rangle$ is an inner product on V, then the pair $(V, \langle ., . \rangle)$ is called an **inner product space**.

Example: Euclidean space

We get an inner product on $\mathbb{R}^n$ by defining, for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}.$$

To verify that this is an inner product, one needs to show that all four properties hold. We check only two of them here.

(i) We have

$$\langle \mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^T \mathbf{x} = x_1^2 + x_2^2 + \cdots + x_n^2 \geq 0,$$

with equality if and only if $x_i = 0$ for all i, that is, $\mathbf{x} = \mathbf{0}$.

(iii) We have

$$\begin{aligned} \langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle &= (\mathbf{x} + \mathbf{y})^T \mathbf{z} = (\mathbf{x}^T + \mathbf{y}^T)\mathbf{z} \\ &= \mathbf{x}^T \mathbf{z} + \mathbf{y}^T \mathbf{z} = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle, \end{aligned}$$

where we have used that matrix multiplication distributes over addition.

Example: Matrix space

We get an inner product on $\mathbb{M}_{m\times n}$ by defining, for $\mathbf{A},\mathbf{B}\in\mathbb{M}_{m\times n}$,

$$\langle \mathbf{A},\mathbf{B}\rangle = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ij}$$

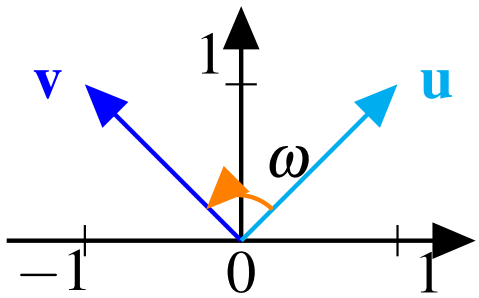
(multiply corresponding entries and add). For instance,

$$\begin{aligned} & \left\langle \begin{pmatrix} 2 & -1 & 3 \\ 5 & 0 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 3 & 8 \\ 0 & 1 & -2 \end{pmatrix} \right\rangle \\ = \quad & (2)(1)+(-1)(3)+(3)(8)+(5)(0)+(0)(1)+(4)(-2) = 15. \end{aligned}$$

This inner product is identical to the dot product on $\mathbb{R}^{mn}$ if an $m\times n$ matrix is viewed as an $mn\times 1$ matrix by stacking its columns.

Orthogonality

Orthogonality generalizes the notion of perpendicularity from Euclidean space. Two vectors $\mathbf{u}$ and $\mathbf{v}$ are said to be **orthogonal** $\Leftrightarrow \langle \mathbf{u}, \mathbf{v} \rangle = 0$. We write $\mathbf{u} \perp \mathbf{v}$ for shorthand.
Additionally, if $\|\mathbf{u}\| = 1 = \|\mathbf{v}\|$ (unit vectors), then $\|\mathbf{u}\|$and$\|\mathbf{v}\|$ are **orthonormal**. The $\mathbf{0}$-vector is orthogonal to every vector in the vector space.



Remark

In what follows, we'll be using the euclidean norm (unless stated otherwise)

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Dot Product

A particular type of inner product, the **scalar/dot product** in $\mathbb{R}^n$, is given by

$$\mathbf{x}^T\mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

In calculus, the "dot product" of two vectors $\vec{x} = (2, -3)$ and $\vec{y} = (5, 1)$ is

$$\vec{x} \cdot \vec{y} = (2, -3) \cdot (5, 1) = (2)(5) + (-3)(1) = 7$$

(multiply corresponding entries and add). In linear algebra we write these same vectors as

$$\mathbf{x} = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} 5 \\ 1 \end{bmatrix},$$

and express the dot product as

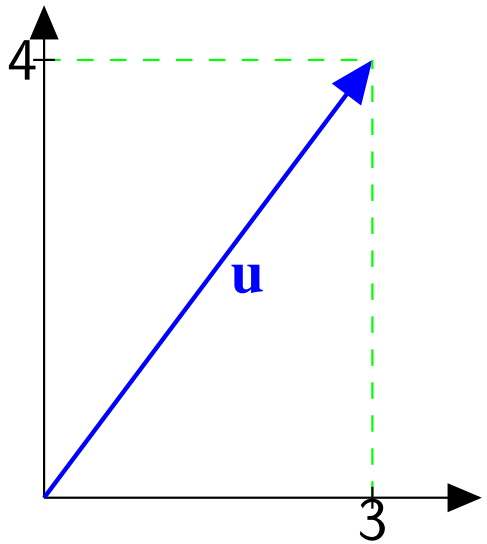
$$\mathbf{x}^T\mathbf{y} = \begin{bmatrix} 2 & -3 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \end{bmatrix} \quad \text{(or just 7)}$$

(so $\vec{x} \cdot \vec{y}$ becomes $\mathbf{x}^T\mathbf{y}$).

Length

The length of a vector $\mathbf{u}$ is denoted $\|\mathbf{u}\|$. A formula for this length comes from the Pythagorean theorem. For instance, if $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, then

$$\|\mathbf{u}\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$



Note that

$$\mathbf{u}^T\mathbf{u} = \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3^2 + 4^2,$$

which is what appears under the square root. In general we have

Distance

The **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ is given by the Euclidean distance formula:

$$d(\mathbf{u},\mathbf{v}) = \|\mathbf{u}-\mathbf{v}\| = \sqrt{\langle \mathbf{u}-\mathbf{v}, \mathbf{u}-\mathbf{v}\rangle}$$

If we use the *dot product* as the inner product, then the distance is called Euclidean distance.

$$d(\mathbf{u},\mathbf{v}) = \|\mathbf{u}-\mathbf{v}\| = \sqrt{(u_1-v_1)^2+(u_2-v_2)^2+\cdots+(u_n-v_n)^2}$$

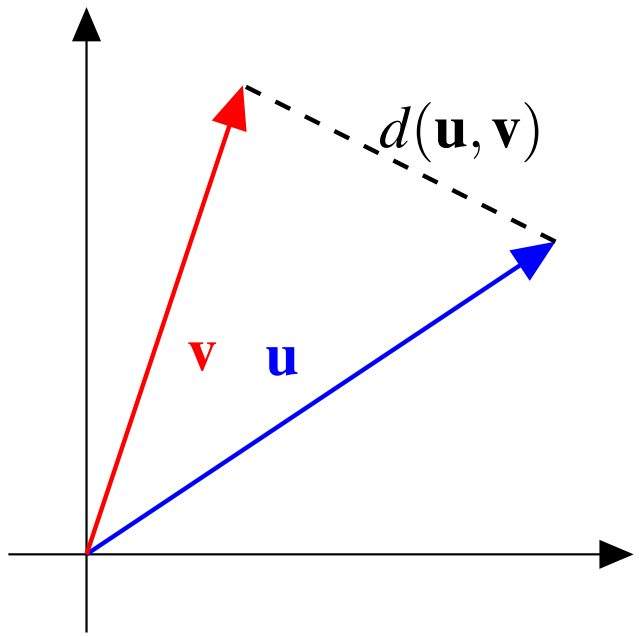
In matrix notation:

$$d(\mathbf{u},\mathbf{v}) = \sqrt{(\mathbf{u}-\mathbf{v})^T(\mathbf{u}-\mathbf{v})}$$

example

Suppose $\mathbf{u} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$. The Euclidean distance between $\mathbf{u}$ and $\mathbf{v}$ is:

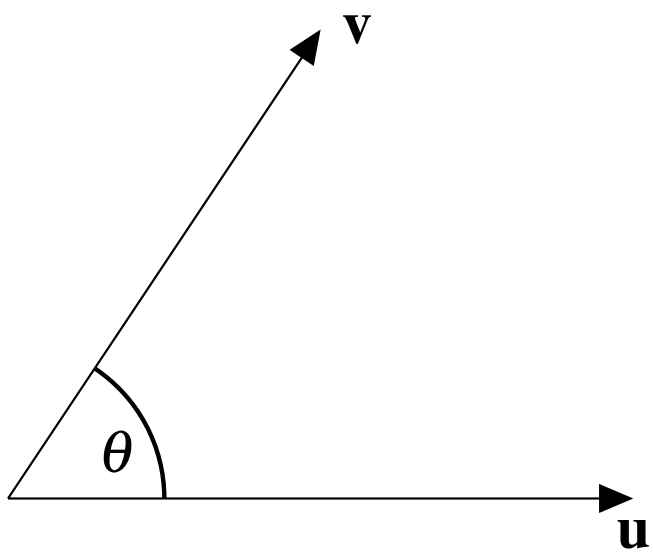
$$d(\mathbf{u},\mathbf{v}) = \sqrt{(5-1)^2 + (3-4)^2} = \sqrt{16+1} = \sqrt{17}$$



Angle

The **angle** θ between two vectors $\mathbf{u}$ and $\mathbf{v}$ is related to the dot product by the formula

$$\mathbf{u}^T\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$$



Remark

The angle between two vectors tells us how similar their orientations are.

Example

Angle between vectors

Let $\mathbf{u} = [2, -3]^T$ and $\mathbf{v} = [3, 2]^T$. The angle between the two vectors is given by:

$$cos(\theta) = \frac{\mathbf{u}^T\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{0}{\sqrt{13}} = 0$$

Hence, $\theta = cos^{-1}(0) = 90°$.
This example shows that

$$\mathbf{u} \perp \mathbf{v} \Leftrightarrow \mathbf{u}^T\mathbf{v} = 0$$

Orthonormal basis

Let V be a vector space and consider a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ (basis) in V.
An **orthonormal basis** is a set of vectors in a vector space that are both orthogonal (mutually perpendicular) and normalized (each vector has a unit length, i.e., a norm of 1). This means:

- The vectors are orthogonal:

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \mathbf{v}_i^T \mathbf{v}_j = 0 \quad \text{for} \quad i \neq j$$

- Each vector has unit length:

$$\|\mathbf{v}_i\| = 1 \quad \text{for all} \quad i$$

Thus, the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ forms an orthonormal basis for the vector space V.

Vector Normalization

Given a vector $\mathbf{v}$ in a vector space V. The normalized version of $\mathbf{v}$, denoted by $\hat{\mathbf{v}}$, is calculated as:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

where $\|\mathbf{v}\|$ represents the norm (or magnitude) of the vector $\mathbf{v}$, calculated as:

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Thus, the normalized vector $\hat{\mathbf{v}}$ has a unit length of 1.

Remark

The concept of an orthonormal basis is critical when discussing support vector machines and principal component analysis.

Example 1: $\mathbb{R}^2$

The canonical/standard basis for a Euclidean vector space $\mathbb{R}^n$ is an orthonormal basis, where the inner product is the dot product of vectors.

Consider the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ in $\mathbb{R}^2$.

1. Orthogonality:The dot product of $\mathbf{v}_1$ and $\mathbf{v}_2$ is:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 1 \times 1 + 1 \times (-1) = 0$$

which shows that the vectors are orthogonal.

2. Normalization: The norm of each vector is:

$$\|\mathbf{v}_1\| = \sqrt{1^2 + 1^2} = \sqrt{2}, \ \|\mathbf{v}_2\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

The normalized vectors are:

$$\hat{\mathbf{v}}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \hat{\mathbf{v}}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The set $\{\hat{\mathbf{v}}_1, \hat{\mathbf{v}}_2\}$ forms an orthonormal basis for $\mathbb{R}^2$.

Example 2: $\mathbb{R}^3$

Consider the vector space $\mathbb{R}^3$. A common example of an orthonormal basis in this space is the standard basis:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

These vectors satisfy the conditions for orthonormality:

1. Orthogonality: The dot product of any two distinct vectors is zero:

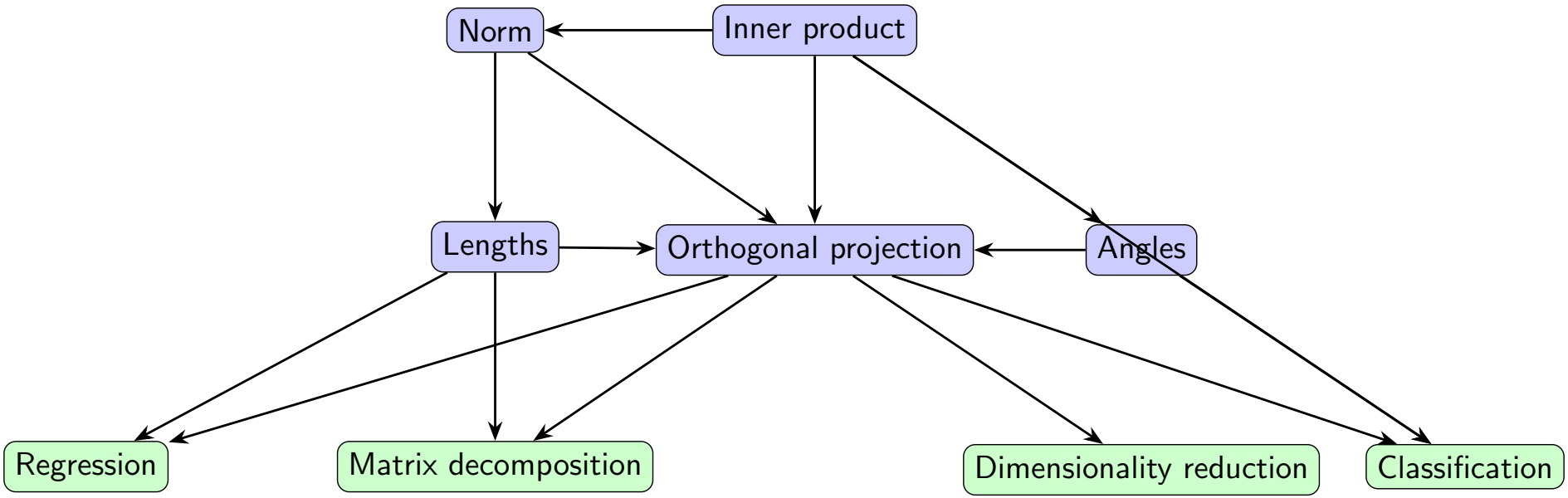
$$\mathbf{e}_i \cdot \mathbf{e}_j = 0 \quad \text{for} \quad i \neq j$$

2. Normalization: Each vector has a unit length:

$$\|\mathbf{e}_i\| = 1 \quad \text{for all} \quad i$$

Therefore, $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ forms an orthonormal basis for $\mathbb{R}^3$.

Put it all together



Applications

- Principal Component Analysis (PCA): *Scalar Product* (to calculate the variance along a particular direction (eigenvector)), *Norm* (to normalize the eigenvectors (principal components) so that they have unit length), *Euclidean distance* (implicitly used when computing the covariance matrix, which captures the spread of data points in the feature space).
- K-Means Clustering: *Euclidean distance* (to calculate the distance between data points and cluster centroids)
- Support vector machines: *Euclidean Norm* (used in the SVM optimization process to maximize the margin, making the classifier more robust),*Scalar Product* (used in defining the decision boundary: $f = wx + b$)
- etc.

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Définition:
La diagonalisation est le processus de transformation d'une matrice carrée A en une matrice diagonale D telle que:

$$A = PDP^{-1}$$

où P est une matrice dont les colonnes sont les vecteurs propres de A et D est une matrice diagonale dont les éléments sont les valeurs propres de A.

Étapes de la Diagonalisation

Pour diagonaliser une matrice A, suivez ces étapes :

1. **Calcul des valeurs propres:** Trouver les valeurs propres λ en résolvant le polynôme caractéristique:

$$\det(A - \lambda I) = 0$$

2. **Calcul des vecteurs propres:** Pour chaque valeur propre λ, trouver les vecteurs propres $\mathbf{v}$ en résolvant:

$$(A - \lambda I)\mathbf{v} = 0$$

3. **Formation de la matrice P:** Construire la matrice P en utilisant les vecteurs propres comme colonnes.
4. **Construction de la matrice diagonale D:** La matrice D est une matrice diagonale dont les éléments sont les valeurs propres de A.

Exemple

Soit la matrice A:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Étape 1: Calculer les valeurs propres:

$$\det(A - \lambda I) = \det \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = 0$$

Résolution de cette équation nous donne $\lambda_1 = 5$ et $\lambda_2 = 2$.

Exemple (suite)

Étape 2: Trouver les vecteurs propres correspondants:

$$(A-5I)\mathbf{v}=0 \quad \text{et} \quad (A-2I)\mathbf{v}=0$$

Pour $\lambda_1=5$, le vecteur propre est $\mathbf{v}_1=\begin{pmatrix}1\\1\end{pmatrix}$.

Pour $\lambda_2=2$, le vecteur propre est $\mathbf{v}_2=\begin{pmatrix}1\\-2\end{pmatrix}$.

Exemple (suite)

Étape 3: Construire la matrice P avec les vecteurs propres comme colonnes:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$$

Étape 4: Construire la matrice diagonale D:

$$D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$$

Conclusion: La matrice A est diagonalizable et peut être exprimée comme $A = PDP^{-1}$.

Conclusion

La diagonalisation simplifie les calculs tels que la puissance de matrices. Elle est essentielle dans diverses applications mathématiques et scientifiques.

$$A^n = PD^nP^{-1}$$

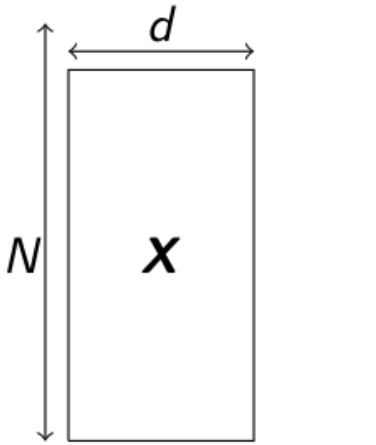
Remarque: Toutes les matrices ne sont pas diagonalizables. Une condition nécessaire est que A ait suffisamment de vecteurs propres linéairement indépendants.

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Réduction de la dimension : le but

Soit $\boldsymbol{X} \in \mathbb{R}^{N \times d}$ (N échantillons de dimension d).

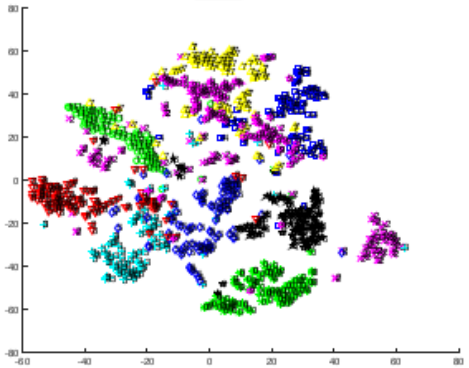
But : trouver une projection de $\boldsymbol{X}$ sur $\boldsymbol{Z} \in \mathbb{R}^{N \times q}$ tel que $q < d$





$d = 784$





$q = 2$

Utilité de la réduction de dimension

Visualization

- $q = 2 \ ou \ 3$
- Explicabilité
- Détecter les valeurs aberrantes

Data representation $q < d$

- Réduire le bruit
- Accélération des calculs
- limiter la malédiction de la dimension (curse of dimensionality)

Utilité de la réduction de dimension

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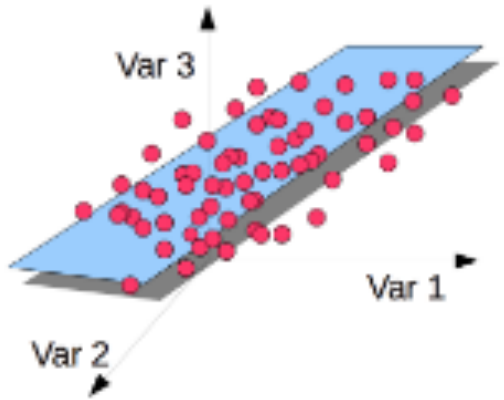
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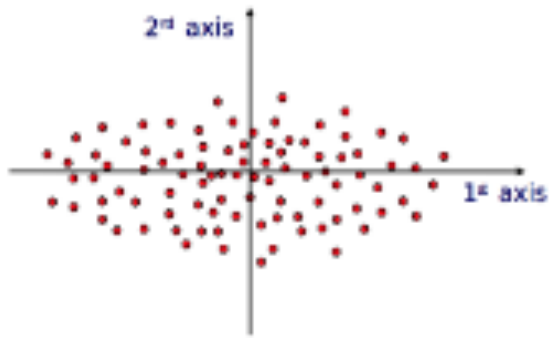
l'Analyse en Composantes Principales

Motivation

Pourquoi ?

- Étudier et visualiser des corrélations entre les variables (limiter le nombre de variables à mesurer)
- Obtenir de facteurs non corrélés qui sont des combinaisons linéaires des variables de départ (pour la modélisation)
- Visualiser des observations dans un espace à deux ou trois dimensions (identifier des groupes)

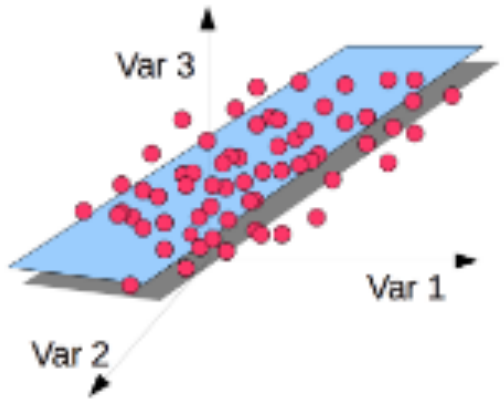


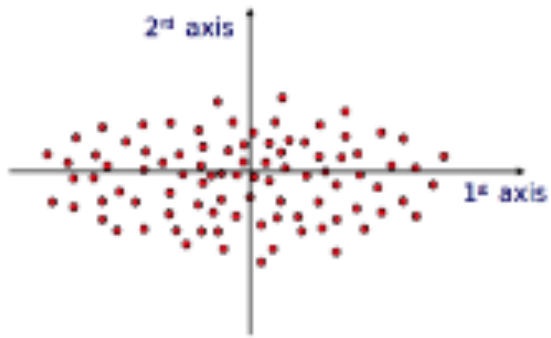


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Comment : Calculer les vecteurs propres de la matrice de covariance ?

Les données : description

Données

Soit $\mathbf{X}$ la matrice des données :

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1j} & \dots & x_{1d} \\ \vdots & & \vdots & \vdots & \vdots & \vdots \\ x_{i1} & x_{i2} & \vdots & x_{ij} & \vdots & x_{1d} \\ \vdots & & \vdots & \vdots & \vdots & \vdots \\ x_{N1} & x_{N2} & \vdots & x_{Nj} & x_{Nd} & \vdots \end{pmatrix} = (X_1, X_2, \dots, X_j, \dots, X_d)$$

Notations

- Observation : $\mathbf{x_i} \in \mathbb{R}^d$ avec $\mathbf{x_i} = (x_{i1}, x_{i2}, \dots, x_{id})^T$
- Variable : X_j
- Moyenne : $\bar{X}_j = \frac{1}{N} \sum_{i=1}^{N} x_{ij}$
- Variance des variables : $\sigma_j^2 = \frac{1}{N} \sum_{i=1}^{N} (x_{ij} - \bar{X}_j)^2$

Covariance

Covariance empirique entre variables j et k

$$\text{cov}(X_j, X_k) = \frac{1}{N}\sum_{i=1}^{N}(x_{ij} - \bar{X}_j)(x_{ik} - \bar{X}_k)$$

Soit $\mathbf{\Sigma} \in \mathbb{R}^{d\times d}$ est une matrice symétrique de terme général $\mathbf{\Sigma}_{jk} = \text{cov}(X_j, X_k)$:

$$\mathbf{\Sigma} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x_i} - \bar{\mathbf{x}})(\mathbf{x_i} - \bar{\mathbf{x}})^T$$

Remarques

- Toujours centrer avant l'ACP
- Réduction (scaling) également si les orders des grandeur sont très différents

Principe de l'ACP

$\{\mathbf{x_i} \in \mathbb{R}^d\}_{i=1}^N$: ensemble de N points décrits par d variables

Projection des points dans un sous-espace de dimension q $(q << d)$ telle que l'information encodé soit maximale

Comment ?

- Projection linéaire de $\mathbf{x_i} \in \mathbb{R}^d$ sur $\mathbf{z_i} \in \mathbb{R}^q$

$$\mathbf{z_i} = \mathbf{P^T x_i} \quad \text{avec} \quad \mathbf{P} = (\mathbf{p_1}, \ldots, \mathbf{p_q}), \quad \mathbf{p_k} \in \mathbb{R}^d$$

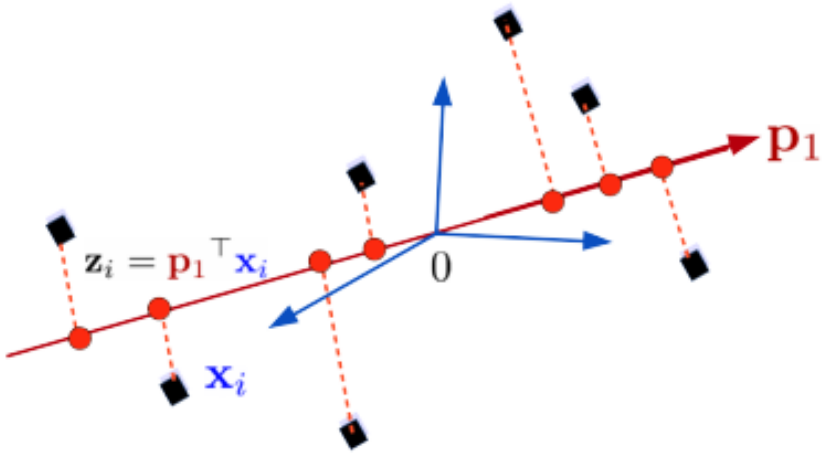
  $\mathbf{P} \in \mathbb{R}^{d \times q}$: matrice de transformation linéaire

- Contrainte : $\mathbf{P^T P = I}$
  Les vecteurs de la nouvelle base sont orthogonaux 2 à 2 :

$$\mathbf{p_j^T p_k} = 0, \forall \text{ k} \neq \text{j}$$

Recherche du premier axe factoriel $\mathbf{p_1}$ de $\mathbf{P}$

- Données: $\{\mathbf{x_i} \in \mathbb{R}^d\}_{i=1}^N$
- $\mathbf{x_i}$ sont normalisés (soustraction de la moyenne et mise à l'échelle de l'écart-type)
- Projections sur $\mathbf{p_1}$: $\{\mathbf{z_i} = \mathbf{p_1^T x_i} \in \mathbb{R}\}_{i=1}^N$



Comment calculer $\mathbf{p_1}$

Nous cherchons $\mathbf{p_1}$ un vecteur unitaire de manière à maximiser la variance des $\mathbf{z_i}$

$$\max_{\mathbf{p_1} \in \mathbb{R}^d} \frac{1}{N}||\mathbf{X}\mathbf{p_1}||_2^2 = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{p_1}^T \mathbf{x_i})^2 \quad \text{s.t.} \quad ||\mathbf{p_1}||^2 = 1$$

$\Longrightarrow$ Cela se transforme en un problème d'optimisation sous contrainte

Variance des points projetés

$J_e(\mathbf{p_1})$: variance de la projection de $\mathbf{x_i}$ sur $\mathbf{p_1}$

$$\begin{aligned} J_e(\mathbf{p_1}) &= \frac{1}{N}\sum_{i=1}^{N} \mathbf{z_i}^2 = \frac{1}{N}\sum_{i=1 in}^{N} \mathbf{p_1^T x_i x_i^T p_1} \\ J_e(\mathbf{p_1}) &= \mathbf{p_1^T \Sigma p_1} \end{aligned}$$

où $\mathbf{\Sigma} = \frac{1}{\mathbf{N}}\mathbf{X^T X}$ est la matrice de corrélation.

Problème d'optimisation

$$\max_{\mathbf{p}_1 \in \mathbb{R}^d} \quad \mathbf{p_1^T \Sigma p_1} \quad \text{tq.} \quad \mathbf{p_1^T p_1} = 1$$

ou de façon équivalente

$$\min_{\mathbf{p}_1 \in \mathbb{R}^d} \quad -\mathbf{p_1^T \Sigma p_1} \quad \text{tq.} \quad \mathbf{p_1^T p_1} = 1$$

Premier axe factoriel : solution

- Lagrangien : $\mathcal{L}(\mathbf{p_1}, \lambda_1) = -\mathbf{p_1^T \Sigma p_1} + \lambda_1(\mathbf{p_1^T p_1} - 1)$
- Conditions d'optimalité :

$$\nabla_{p_1} \mathcal{L} = -2\mathbf{\Sigma p_1} + 2\lambda_1 \mathbf{p_1} = 0 \quad \text{et} \quad \nabla_{\lambda_1} \mathcal{L} = \mathbf{p_1^T p_1} - 1 = 0$$

- Solutions :

$$\mathbf{\Sigma p_1} = \lambda_1 \mathbf{p_1} \text{ et } \mathbf{p_1^T \Sigma p_1} = \lambda_1$$

Interprétation

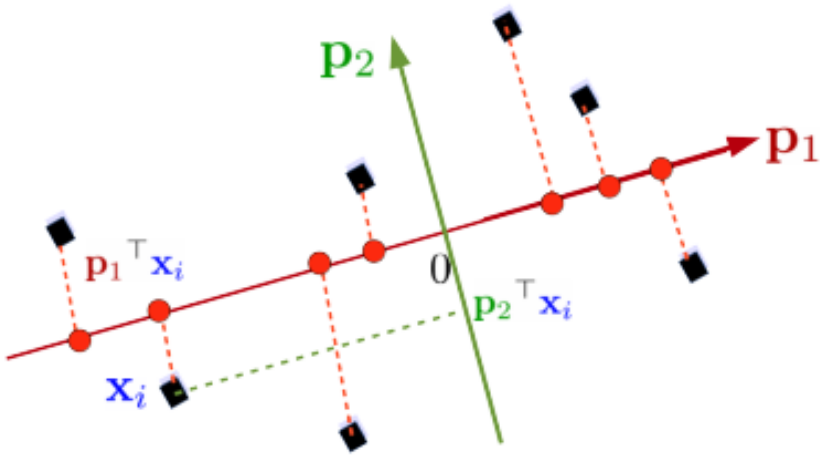
1. $(\lambda_1, \mathbf{p_1})$ est une paire (valeur propre, vecteur propre) de $\mathbf{\Sigma}$
2. $\mathbf{p_1^T \Sigma p_1} = \lambda_1$ est la variance des projections
3. λ_1 doit être maximal
4. $\mathbf{p_1}$ est le vecteur propre associé à la plus grande valeur propre de $\mathbf{\Sigma}$

k^{th} factorial axis

Computing the 2nd factorial axis $\mathbf{p_2}$

$$\min_{\mathbf{p_2} \in \mathbb{R}^d} \quad -\mathbf{p_2^T \Sigma p_2} \quad \text{tq.} \quad \mathbf{p_2^T p_2} = 1, \ \mathbf{p_2^T p_1} = 0$$

we are looking for a unit vector $\mathbf{p_2}$ orthogonal to the vector $\mathbf{p_1}$, that maximizes the variance $\mathbf{p_2^T \Sigma p_2}$



Solution :

- $\mathbf{p_2}$ is the eigenvector associated with λ_2, the second greatest eigenvalue of $\mathbf{\Sigma}$
- $\mathbf{p}_k$ is the eigenvector associated with the kth highest eigenvalue λ_k of $\mathbf{\Sigma}$ (orthogonal to all first $k-1$ axes)

How does it work?

1. Normalize the data $\{\boldsymbol{x_i} \in \mathbb{R}^d\}_{i=1}^N \longrightarrow \{x_{ij} = \frac{x_{ij}-\bar{x}_j}{\sigma_j}, j = 1, \ldots, d\}_{i=1}^N$
2. Calculate the correlation matrix $\boldsymbol{\Sigma} = \frac{1}{N} X^T X$
3. Compute the eigenvalue decomposition $\{\mathbf{p_j} \in \mathbb{R}^d, \lambda_j \in \mathbb{R}\}_{j=1}^d$ of $\boldsymbol{\Sigma}$
4. Sort the eigenvalues λ_j in decreasing order
5. New basis for data representation

$$\mathbf{P} = (\mathbf{p_1}, \ldots, \mathbf{p_q}) \in \mathbb{R}^{d \times q}$$

where $(\mathbf{p_1}, \ldots, \mathbf{p_q})$ are the q eigenvectors associated with the q highest eigenvalues.

6. Projection of all points onto $\mathbf{P}$

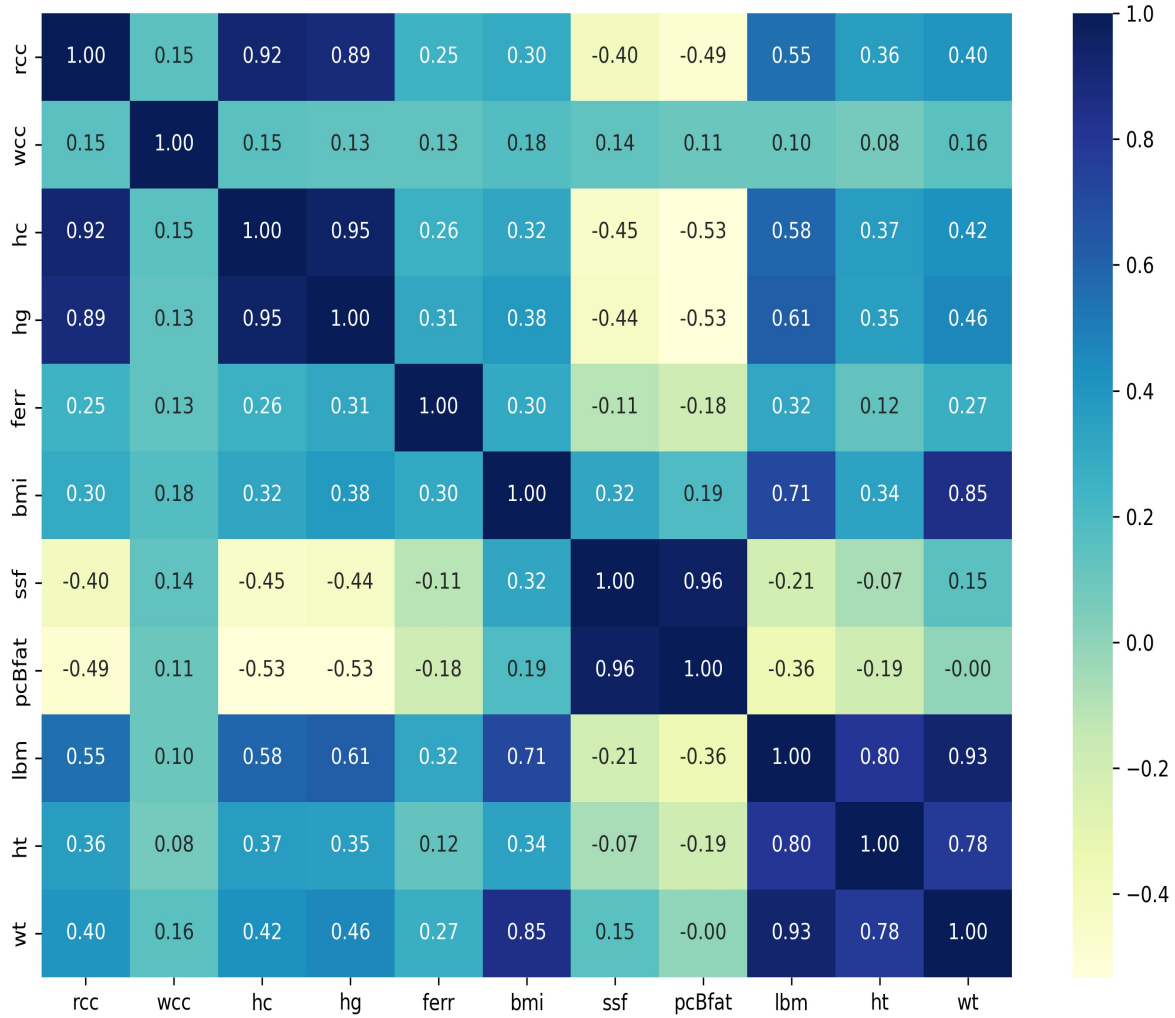
$$\mathbf{Z^T} = (\mathbf{z_1}, \ldots, \mathbf{z_N}) \quad \text{with} \quad \mathbf{z_i} = \mathbf{P^T x_i}$$

Athlete's dataset

Objectif : explorer la corrélation entre les variables (mesures physiologiques) et visualiser la représentation 2d des athlètes.

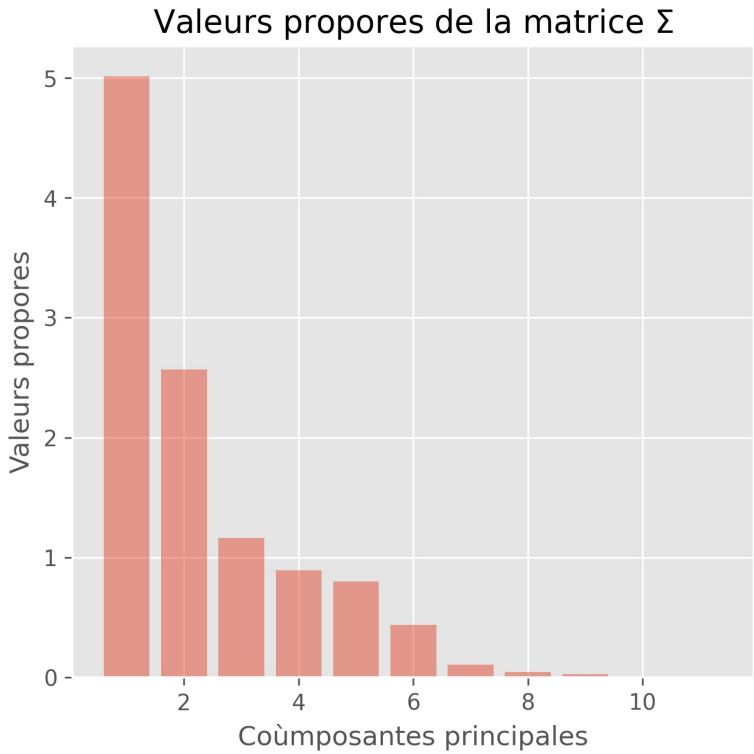
rcc	wcc	hc	hg	ferr	bmi	ssf	pcBfat	lbm	ht	wt	sex
4.82	7.6	43.2	14.4	58	22.37	50	11.64	53.11	163.9	60.1	f
4.32	6.8	40.6	13.7	46	17.54	54.6	12.16	46.12	173	52.5	f
5.16	7.2	44.3	14.5	88	18.29	61.9	12.92	48.76	175	56	f
4.66	6.4	40.9	13.9	109	18.37	38.2	8.45	41.93	157.9	45.8	f
4.19	9	39	13.4	69	18.93	43.5	10.16	42.95	158.9	47.8	f
4.53	5	40.7	14	41	17.79	56.8	12.55	38.3	156.9	43.8	f
4.42	6.4	42.8	14.5	63	20.31	58.9	13.46	39.03	149	45.1	f
4.32	4.3	41.6	14	177	26.73	35.2	6.46	91	190.4	96.9	m
4.73	6.7	42.8	14.9	8	19.81	41.8	7.19	70	195.2	75.5	m
4.71	7.2	43.6	14	32	20.39	30.5	5.63	67	186.6	71	m
4.93	7.3	46.2	15.1	41	21.12	34	6.59	67	184.4	71.8	m
5.21	7.5	47.5	16.5	20	21.89	46.7	9.5	70	187.3	76.8	m

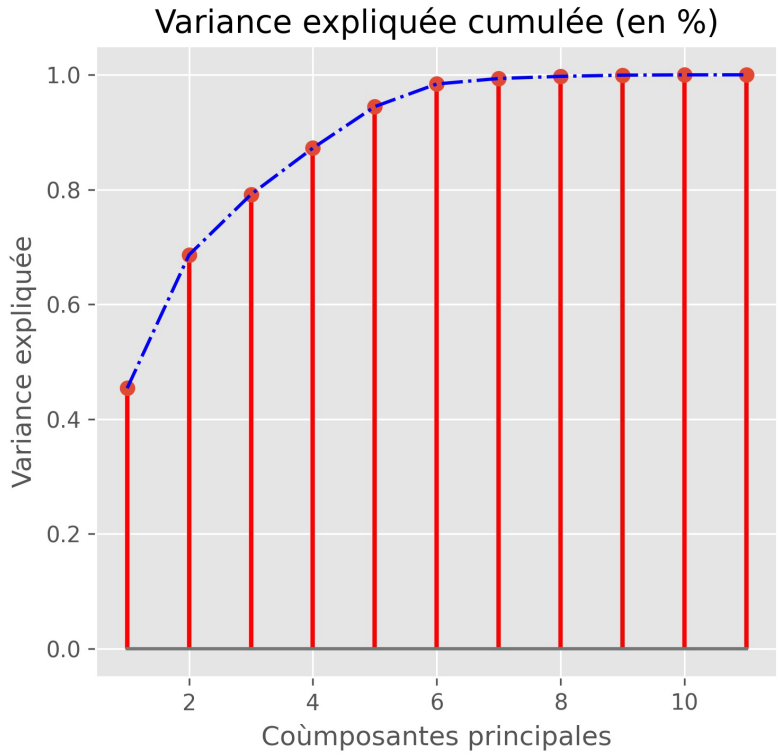
Corrélation



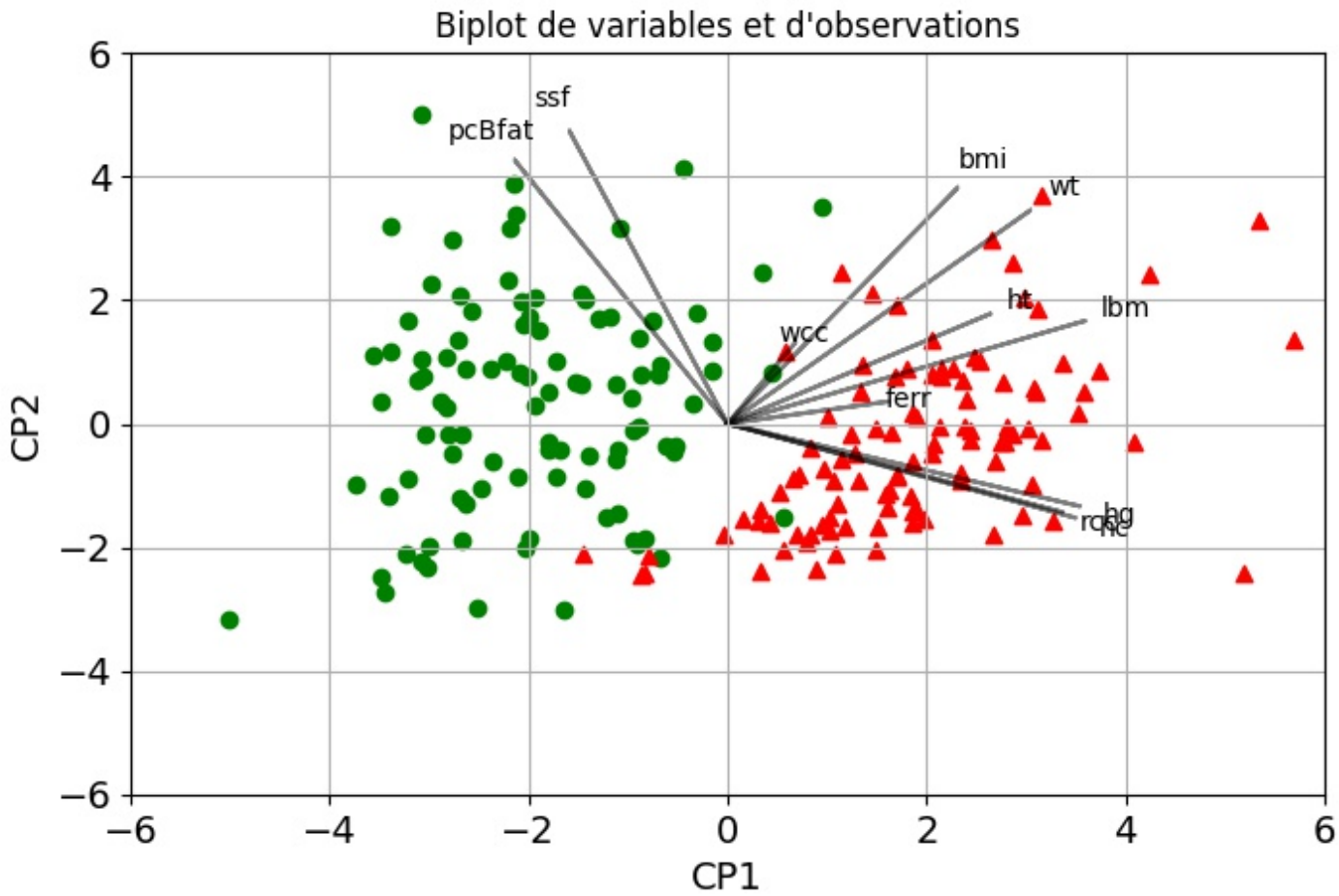
How to chose q?

- Detect "an elbow" on the graph of eigenvalues
- We choose q s.t. a fixed percentage (for instance 95%) of the variance is kept
- Select the principal components with an eigenvalue greater than 1 ($\lambda_j < 1$).

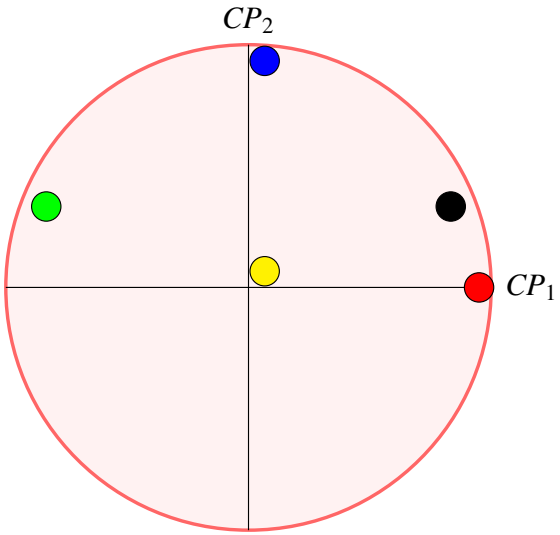




Analyse des résultats



Le cercle



Deux variables sont corrélées si elles sont proches les unes des autres et de la circonférence du cercle de corrélation. Chaque point sur le cercle représente une variable.

- Les points noir et rouge sont corrélés parce qu'ils sont proches les uns des autres et de la circonférence du cercle.
- Le point rouge est positionner sur l'axe CP_1. Cela indique que cette variable est parfaitement corrélée avec lui.
- Le point bleu chevauche presque l'axe CP_2. Cela indique que cette variable est fortement corrélée avec lui.

- Le point jaune est loin de la circonférence du cercle. Il se trouve proche du centre. Ceci signifie que la corrélation est très faible.
- L'angle entre les points rouge et bleu est presque de $90°$, donc ces points ne sont pas corrélés.
- La corrélation entre les points vert et rouge est forte, mais elle est négative (Nous distinguons la puissance de la corrélation et son signe. Une corrélation forte mais négative indique que deux variables sont inversement proportionnelles. Si l'une augmente, l'autre diminue).

Propriétés des axes factoriels

Pro1 Les valeurs propres de $\mathbf{\Sigma}$ sont positives car $\mathbf{\Sigma}$ est une matrice semi-definie positive

Pro2 Le nombre d'axes factoriels est égal au nombre de valeurs propres non-nulles de $\mathbf{\Sigma}$

Pro3 Variance expliqué par l'axe factoriel $\mathbf{p_k}$: $I_k = \mathbf{p_k^T \Sigma p_k} = \lambda_k$

Pro4 Variance totale des axes factoriels $I = \sum_{k=1}^{q} \lambda_k =$ trace of $\mathbf{\Sigma}$

Pro5 Pourcentage de variance expliquée par les q premiers axes

$$\frac{\sum_{k=1}^{q} \lambda_k}{\sum_{k=1}^{d} \lambda_k} \times 100$$

Conclusion

- PCA $\equiv$ represent data in a reduced-dimensional space
- The principal components $\mathbf{z_i}$ $\forall j = 1, \ldots, d$ are centered and uncorrelated $\text{cov}(\mathbf{z_j}, \mathbf{z_k}) = 0 \quad \forall j \neq k$
- Let $\mathbf{z_k} = \mathbf{X}\mathbf{p_k}$ be the projection of $\mathbf{X}$ on the $k-th$ axis $\mathbf{p_k}$. The variance of the principal component $\mathbf{z_k}$ is $cov(\mathbf{z_k}, \mathbf{z_k}) = \lambda_k$
- Reconstructing $\mathbf{x_i}$ using the principal components :

$$\hat{\mathbf{x}}_\mathbf{i} = \mathbf{P}\mathbf{z_i}$$

- The principal components are linear combinations of the original variables.

Drawbacks

- PCA : linear projection method
- Implicit hypothesis : data follows a gaussian distribution